Year 7
Mathematics

Unit 3: Equations, Functions and Formulae

3.2 Writing Algebraic Expressions

Writing Algebraic Expressions

STARTER · 5 MIN

Date: _____ **Title: Writing Algebraic Expressions**

LEARNING OBJECTIVES

- To be able to write expressions using the four operations

STARTER *Answer in your book. No calculators.*

1. Simplify $4x + 3x$
2. Simplify $9a - a$
3. Simplify $2b + 5 + b$
4. Simplify $y \times y$

Starter — Answers

STARTER

Tick or correct in green pen

1. $7x$

2. $8a$

3. $3b + 5$

4. y^2

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write expressions using one operation	☐
SILVER	I can write expressions using two operations in the right order	☐
GOLD	I can use function machines to write expressions and work backwards	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Writing Algebraic Expressions

INTRODUCTION · 3 MIN

A letter stands for an unknown number. Write × and ÷ the algebra way.

Add / subtract

n more than ... / ... less than

5 more than *n*: $n + 5$

Multiply

Write the number first, no × sign.

3 times *n*: $3n$

Divide

Write it as a fraction.

n divided by 4: $\frac{n}{4}$

Order matters

'Subtract 3 from *n*'

$n - 3$, not $3 - n$

Examples 1 and 2 — Words into Algebra

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint 3 times n = $3n$ $n \div 4$ = fraction

Example 1

Write an expression for

(a) 4 more than x

(b) y take away 7

Example 2

Write an expression for

(a) 5 times n

(b) a divided by 3

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Add or subtract

(a) $x + 4$

(b) $y - 7$

Tip: 'take away 7' means − 7.

Example 2

Number first, divide as a fraction

(a) $5n$

(b) $\frac{a}{3}$

Tip: never write $n5$.

Task 1

PRACTICE · 9 MIN

BRONZE *Write an expression*

Hint 3 times n = $3n$ $n \div 4$ = fraction

1. 6 more than m
2. p minus 2
3. 8 times k
4. t divided by 5
5. 3 less than x

★ y multiplied by 10

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $m + 6$

2 $p - 2$

3 $8k$

4 $\frac{t}{5}$

5 $x - 3$

★ $10y$

Examples 3 and 4 — Two Operations

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Do the operations in order Simplify the total

Example 3

Write an expression for 'multiply n by 3, then add 2'

Example 4

Ben has b sweets.
He eats 4, then shares the rest equally with a friend.

How many does each get?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Do the operations in order

× 3 $= 3n$

then + 2 $= 3n + 2$

Tip: $3n + 2$ is not $3(n + 2)$.

Example 4

Subtract first, then divide

Eats 4 $= b - 4$

Each gets $= \dfrac{b - 4}{2}$

Tip: the fraction line acts as a bracket.

Task 2

PRACTICE · 9 MIN

SILVER *Write an expression*

Hint Do the operations in order Simplify the total

1. Multiply x by 4, then add 1
2. Add 5 to y, then multiply by 2
3. Double a, then subtract 3
4. Divide m by 2, then add 6
5. A pen costs p pence.
   Cost of 6 pens?

★ Amy is a years old. Her brother is 3 years older.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $4x + 1$

2 $2(y + 5)$

3 $2a - 3$

4 $\frac{m}{2} + 6$

5 $6p$ pence

★ $a + 3$

Examples 5 and 6 — Function Machines

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

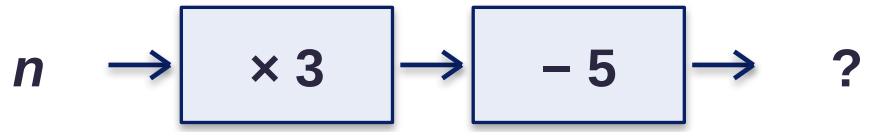
Hint Work backwards: inverses + first: use a bracket

Example 5

(2 marks)

Write the output for input n.

Then find the output when $n = 5$.

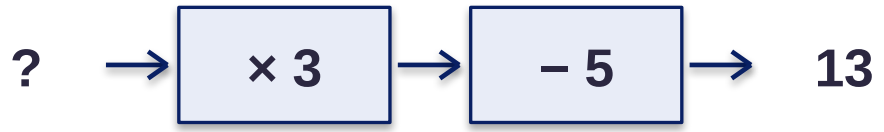


Example 6

(2 marks)

The output is 13.

Work backwards to find the input.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Apply the operations in order

Output $= 3n - 5$

$n = 5 \quad = 3 \times 5 - 5$

$\quad\quad\quad\quad = \mathbf{10}$

Tip: × 3 first, then − 5.

Example 6

Use the inverse operations

$13 + 5 = 18$

$18 \div 3 = 6$

Input = 6

Tip: undo − 5 first, then undo × 3.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: function machines*

Hint Work backwards: inverses + first: use a bracket

1 $n \rightarrow \times 2 \rightarrow + 7$

Output for input n?

2 $n \rightarrow + 4 \rightarrow \times 3$

Output for input n?

3 $n \rightarrow \times 5 \rightarrow - 1$

Output when $n = 4$?

4 $n \rightarrow \times 2 \rightarrow + 7$

Input if the output is 19?

5 $n \rightarrow \div 2 \rightarrow + 3$

Output for input n?

★ $n \rightarrow \times 4 \rightarrow - 6$

Input if the output is 30?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 $2n + 7$

2 $3(n + 4)$

3 $5 \times 4 - 1 = 19$

4 $(19 - 7) \div 2 = 6$

5 $\frac{n}{2} + 3$

★ $(30 + 6) \div 4 = 9$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write an expression for 7 less than x

SILVER Write y divided by 4 the algebra way

GOLD $n \rightarrow \times 3 \rightarrow -5$. What is the output for input n?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x - 7$

SILVER $\frac{y}{4}$

GOLD $3n - 5$

Exam Practice — Writing Expressions

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint 3 times n = $3n$ $n \div 4$ = fraction

Question 1

(2 marks)

Write an expression for

(a) 8 more than x (1)

.....

(b) y multiplied by 6 (1)

.....

Question 2

(2 marks)

Write an expression for

(a) m divided by 3 (1)

.....

(b) 5 less than k (1)

.....

Exam Practice — Expressions in Context

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Do the operations in order Simplify the total

Question 3

(2 marks)

A cinema ticket costs t dollars.
Popcorn costs 4 dollars.

Write an expression for the cost of 3 tickets and one popcorn.

.....

Question 4

(2 marks)

Ali has a stickers. Ben has twice as many as Ali.

Write an expression for the total number of stickers.

.....

Exam Practice — Function Machines

EXAM PRACTICE · 10 MIN

GOLD

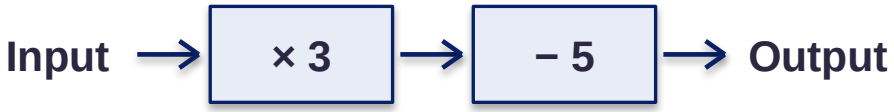
IGCSE exam-style: show all your working

Hint Work backwards: inverses + first: use a bracket

Question 5

(3 marks)

Here is a function machine.



(a) Output when the input is 7? (1)

.....

(b) Input when the output is 25? (1)

.....

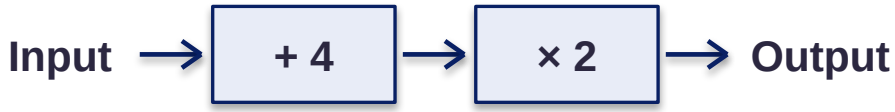
(c) Output for input n? (1)

.....

Question 6

(2 marks)

Here is a function machine.



(a) Output for input n? (1)

.....

(b) Input when the output is 30? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $x + 8$ — B1

(b) $6y$ — B1

Tip: number first: 6*y*.

Question 2

(a) $\frac{m}{3}$ — B1

(b) $k - 5$ — B1

Tip: 5 less than *k* is *k* − 5.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

3 tickets: $3t$ — M1

$3t + 4$ — A1

Tip: multiply first, then add.

Question 4

Ben: $2a$ — M1

$a + 2a = 3a$ — A1

Tip: simplify the total.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $7 \times 3 - 5 = 16$ — B1

(b) $(25 + 5) \div 3 = 10$ — B1

(c) $3n - 5$ — B1

Tip: work backwards with inverses.

Question 6

(a) $2(n + 4)$ — B1

(b) $30 \div 2 - 4 = 11$ — B1

Tip: + 4 first, so use a bracket.